1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L36 – Project Management**

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4 **Recap**

- Requirement stage – functional and usability (CI)
- Interface design – design guidelines (usability), DFD+ER, OOD+UML
- Prototype (including evaluation) – CW, HE
- System design –DFD+ER, OOD+UML
- Coding
- Code testing – review-based, black-box, white-box, integration + system testing
- Usability testing

5 **Deployment & Maintenance**

- Important (not much to discuss)
- Maintenance – includes updates, bug-fixes etc

6 **Project Management**

7 **Why**

- Important topic – particularly for large projects
- Prior estimation of manpower, resources and cost preferable

8 **Project Planning**

- Involves estimation of
  - Project size
  - Cost
  - Duration
  - Effort

9 **Project Planning**

- Based on such estimations, a project manager needs to
  - Create staff organization and staffing plan
  - Plan/arrange for other resources
  - Identify risks and plan for mitigation
  - Make plan for quality assurance (QA)
  - …

10 **Project Size Estimation**

- Lines of Code (LoC) – total number of lines of instructions (excluding comments)
  - Difficult to estimate at the beginning
  - Modular hierarchy can help – coupled with prior experience

11 **Project Size Estimation**

- Function Point (FP) metric
  - Project size related to the number of functions supported by the system (along with some other things)

12 **COCOMO**

- Stands for Constructive Cost Estimation Model
- Proposed by Boehm [1981]
- Can be used for cost, duration and effort estimation

13 **COCOMO**

- According to Boehm, software cost estimation should be done through three stages
  - Basic COCOMO
  - Intermediate COCOMO
  - Complete COCOMO

14 **Basic COCOMO**

- Gives estimate with following equations
  - Effort = a x $(KLoC)^{a1}$ PM [PM = person months, KLoC = Kilo LoC]
  - Tdev = b x $(Effort)^{b1}$ Months
- Constants: a, a1, b, b1
- Takes different values for different types of projects

15 **Project Types & Constant Values**

- Organic: if project deals with well understood application program, team size is reasonably small with experienced team members [a=2.4, a1=1.05, b=2.5, b1=0.38]
- Semidetached: team consists of a mixture of experienced and inexperienced staff [a=3.0, a1=1.12, b=2.5, b1=0.35]
- Embedded: if software strongly coupled to complex hardware, or if stringent regulations on operational procedures exist [a=3.6, a1=1.2, b=2.5, b1=0.32]

16 **Scheduling**

- Identify tasks and subtasks
- Determine dependencies (activity network)
- Allocate resources (Gantt chart)
- Time schedule (PERT charts)

17 **Task Division (Hierarchy)**

18 **Activity Dependency (Activity Network)**

19 **Critical Path Analysis**

20 **Gantt Chart**

- A special type of bar chart (named after its developer Henry Gantt) where each bar represents an activity
  - Bar length proportional to duration planned for activity

21 **PERT (Project Evaluation and Review Technique) Chart**

- Consists of a network of boxes and arrows
- Boxes represent activities and arrows represent task dependencies
- Boxes usually annotated with pessimistic, likely, and optimistic estimates for every task

22 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 3